

# Monte Carlo Results: LQT Path-Following Controller

September 22, 2026

## 1 Summary

Two Monte Carlo studies were run on the closed-loop LQT path-following controller. Each trial uses a new random smooth path (15–30 m long,  $v_c = 0.5$  m/s).

1. **Nominal plant** ( $N = 1000$ ): the plant exactly matches the model used by the controller.
2. **Parameter mismatch** ( $N = 500$ ): the plant’s mass, yaw inertia, rolling resistance, sliding friction, wheel radius, track width, and CG position are randomly perturbed, while the controller keeps the nominal values.

**Main finding.** With a perfect model the tracking error is at the millimeter level. With realistic parameter mismatch it grows to tens of centimeters typically and up to about 2 m in the worst trials, and the figures point to one dominant cause: **rolling-resistance ( $C_{rr}$ ) mismatch**. The other six perturbed parameters show no visible effect on final position error. The rest of this report supports that with the figures and suggests two changes to the controller.

**A note on the numbers.** The console statistics were not saved, so all values below are read off the histograms and are approximate (roughly two significant figures). The two studies also use slightly different error metrics (see Section 4), which does not matter for the comparison because the nominal errors are so small.

| Metric                | Nominal ( $N = 1000$ )     | Mismatch ( $N = 500$ )                                                        |
|-----------------------|----------------------------|-------------------------------------------------------------------------------|
| Max cross-track error | 0.4–1.3 mm (typ. 0.9 mm)   | median $\approx 0.3$ m, 95th $\approx 1.5$ m, max $\approx 2.2$ m             |
| Max heading error     | 0.006–0.022° (typ. 0.014°) | median $\approx 5^\circ$ , 95th $\approx 14^\circ$ , max $\approx 18.5^\circ$ |
| Final position error  | $\leq 4$ mm (typ. 1.5 mm)  | median $\approx 0.85$ m, 95th $\approx 1.8$ m, max $\approx 3$ m              |

Table 1: Approximate error distributions read from the histograms. Roughly a quarter of the mismatch trials strayed more than about 1 m from the path.

## 2 Nominal plant

All 1000 simulated paths lie on top of their references (Figure 1). The maximum cross-track error is around a millimeter and the maximum heading error around  $0.01^\circ$  in every trial (Figure 2). The final position error is at most about 4 mm, with a small secondary group of trials near 2.5–3.5 mm.

These numbers are optimistic by construction: the simulated plant is the model the feedforward inverts, so the feedforward nearly tracks the path on its own. This study confirms that the reference generation, feedforward, and sign conventions are consistent across many path shapes. It does not indicate real-hardware accuracy.

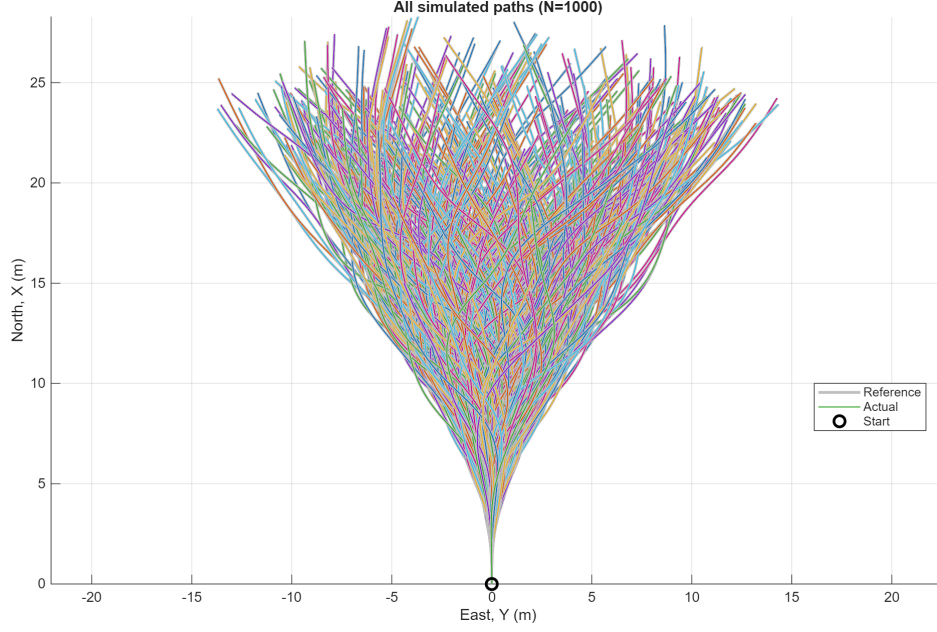


Figure 1: All 1000 nominal-plant trials

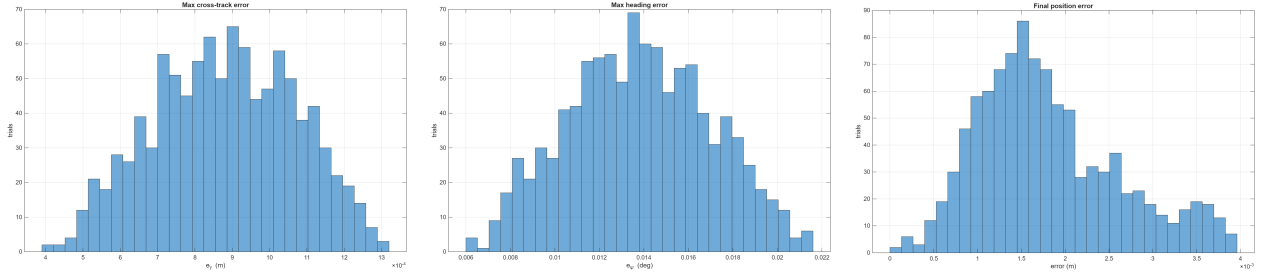


Figure 2: Nominal plant. Left to right: max cross-track error (order  $10^{-3}$  m), max heading error (order  $10^{-2}$  deg), and final position error (order  $10^{-3}$  m).

### 3 Parameter mismatch

#### 3.1 Setup

Plant parameters were drawn independently and uniformly within the ranges below; the controller (feedforward and gains) kept the nominal values.

| Parameter                   | Nominal              | Perturbation       |
|-----------------------------|----------------------|--------------------|
| Mass $m$                    | 20 kg                | $\pm 15\%$         |
| Yaw inertia $I_z$           | $2.0 \text{ kg m}^2$ | $\pm 25\%$         |
| Rolling resistance $C_{rr}$ | 0.02                 | $\pm 50\%$         |
| Sliding friction $\mu$      | 0.6                  | $\pm 30\%$         |
| Wheel radius $r$            | 0.10 m               | $\pm 3\%$          |
| Track width $d$             | 0.25 m               | $\pm 15\%$         |
| CG shift (front/rear)       | 0                    | $\pm 3 \text{ cm}$ |

Table 2: Uncertainty ranges used in the parameter-mismatch study.

### 3.2 Results

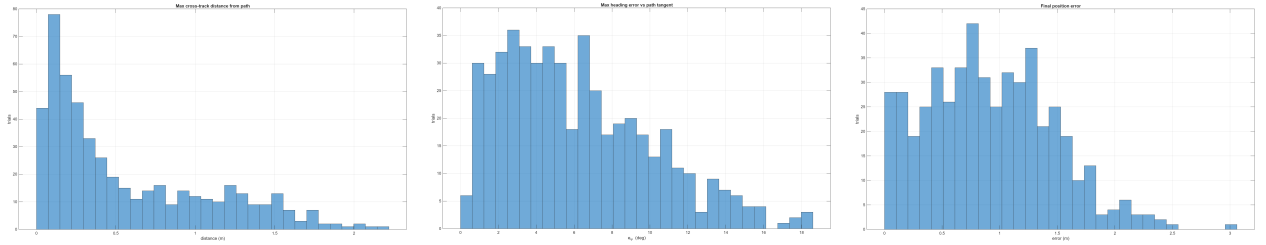


Figure 3: Parameter mismatch. Left to right: max distance from the path, max heading error against the path tangent, and final position error. All three are right-skewed with long tails.

The error distributions (Figure 3) are three orders of magnitude larger than the nominal case. Cross-track distance is concentrated below about 0.5 m (roughly 60% of trials) but has a long tail to about 2 m. Heading error against the path tangent is typically a few degrees, with a tail reaching about  $18^\circ$ . Final position error is spread between 0 and about 2 m, with one outlier near 3 m. For paths that are only 15–30 m long, errors of 1–2 m are large.

### 3.3 Which parameter matters

Figure 4 plots final position error against each perturbation. Only rolling resistance shows structure:

- Final error is close to zero when  $C_{rr}$  is near nominal and grows roughly linearly with  $|\Delta C_{rr}|$  on *both* sides, reaching about 1.5–2.5 m at  $\pm 50\%$ . The V shape means an underestimated  $C_{rr}$  and an overestimated  $C_{rr}$  hurt about equally.
- Mass,  $I_z$ ,  $\mu$ , wheel radius, track width, and CG shift show no visible trend; each is a uniform cloud. (Mass has a weak indirect role, since the rolling-resistance force is  $C_{rr}mg$ .)

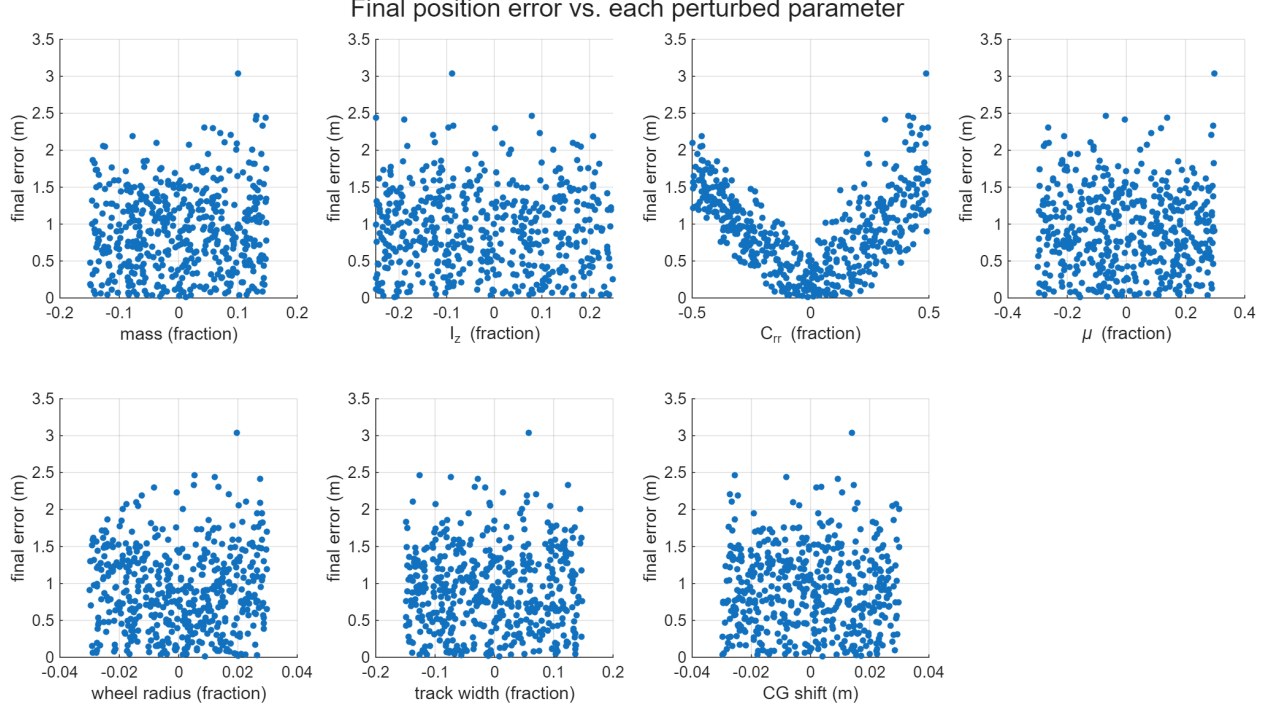


Figure 4: Final position error against each perturbed parameter. Only  $C_{rr}$  shows a clear (V-shaped) dependence.

### 3.4 Why rolling resistance dominates

The feedforward cancels rolling resistance using the nominal  $C_{rr}$ . If the true value differs, the drive torque is off by a constant, and the speed loop is proportional-only, so it settles to a nonzero steady speed error

$$e_v = \frac{r \Delta F_{rr}}{K_p}, \quad \Delta F_{rr} = \Delta C_{rr} mg. \quad (1)$$

For a 50% error ( $\Delta C_{rr} = 0.01$ ),  $\Delta F_{rr} \approx 2$  N and, with  $K_p = 4.8$ ,  $e_v \approx 0.04$  m/s, about 8% of cruise speed. Held for a  $\sim 35$  s cruise, that is roughly 1.4 m of accumulated along-path error, consistent with the 1.5–2.5 m seen at the extremes of Figure 4. (This is a back-of-the-envelope check, not a fit.) The controller has no integral action on speed and no feedback on progress along the path, so nothing removes this drift.

**Hypothesis for the cross-track and heading tails (not directly confirmed by these figures).** The reference is indexed by time, so a rover that has fallen behind schedule is being steered toward a reference point that is ahead of it. On curved sections that point is not on the local path, which would pull the rover off the path and produce the multi-degree heading errors and up-to-2 m path distances seen in Figure 3. Checking this would need path distance plotted against  $C_{rr}$ , which the current figures do not include.

## 4 Metric definitions

- *Nominal study*: cross-track and heading errors are measured against the time-indexed reference point.
- *Mismatch study*: distance to the path and heading error are measured against the *nearest point on the path*, because a lagging rover would otherwise be charged for schedule lag as cross-track error.
- Final position error is the distance from the rover's final position to the path's end point in both studies.

## 5 Recommendations

1. **Add integral action to the speed loop** (or a disturbance observer that estimates  $C_{rr}$  online). This directly removes the steady speed error identified above and should eliminate most of the final-position error.
2. **Make the reference progress-aware.** Project the rover onto the path and advance the reference from the rover's actual progress (with a catch-up term), rather than from time alone, so schedule lag cannot cause cross-track excursions.
3. **Re-run the mismatch study** after each change, and consider a  $C_{rr}$ -only sweep to isolate the effect.
4. **Identify  $C_{rr}$  (and  $\mu$ ) on the real terrain.** These are currently estimates, and  $C_{rr}$  is the parameter the controller is most sensitive to.
5. **Next studies:** sensor noise and latency, actuator saturation, and initial-condition dispersion, as discussed earlier.